How Much of an Expected Goals Model Is in the Words?

Recovering shot quality from free-text match commentary

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Abstract. Expected goals (xG) models are estimated from shot coordinates, which are collected by human annotators or tracking cameras and are, outside a small number of open releases, sold rather than published. Free-text match commentary is subject to neither constraint: it is available without a key, it is published for many competitions that have no open coordinate data, and it describes each shot in a single sentence. We ask how much of a coordinate-based xG model can be recovered when that sentence is the only input.

We parse 87,980 shots from ESPN commentary across six European competitions into eighteen binary and ordinal fields, estimate a logistic regression on three Premier League seasons, and evaluate it on a held-out fourth, obtaining an AUC of 0.7709. To compare words against coordinates directly, we join 8,825 shots to StatsBomb's open Premier League 2015/16 release, for which both a commentary sentence and a hand-collected coordinate xG exist for the same shot. The text model attains an AUC of 0.7826 against StatsBomb's 0.8118, recovering 90.6% of the coordinate model's discrimination above chance. Aggregated to the team-match, the two estimates correlate at $r = 0.869$, which falls within the range reported between commercial providers. Substituting the full sentence for the extracted fields reduces AUC by 0.010, indicating that the extraction is not the binding constraint and that the residual gap reflects information the sentence does not contain.

We then characterise a failure mode particular to text-derived features. Consistent with Robberechts and Davis, a decade-stale model loses only 0.0068 AUC and transfers between leagues without cost, so the induced ranking is stable. The predicted level is not: the model over-estimates the held-out season by 11.9%, which decomposes into a 5.7% decline in the league goal rate and a 5.9% shift in the distribution of parsed phrases. Two phrases account for the second component, while unambiguous phrases such as "penalty" are unchanged. Re-estimation on more recent data does not correct this, whereas refitting a single intercept on shots already played reduces the error to 2.9% while leaving the ranking unaltered. The instability is confined to phrases whose application requires a judgement by the annotator.

Keywords: Expected goals · Football analytics · Text as features · Concept drift · Calibration

1 Introduction

An expected goals model estimates the probability that a given shot results in a goal. Every widely used specification is built on the location from which the shot was taken: a coordinate pair, a derived distance and angle, and increasingly the positions of the other players in frame [6,3]. That location is the expensive component. It is produced either by human annotators or by tracking cameras, and with the exception of a small number of open releases it is licensed rather than published.

Free-text commentary carries none of that cost. For each shot, a sentence of the following form is published live and without authentication:

> *"Attempt saved. Bukayo Saka (Arsenal) right footed shot from the centre of the box is saved in the bottom left corner. Assisted by Martin Ødegaard with a through ball following a fast break."*

Such sentences are available for many competitions for which open coordinate data does not exist and is unlikely to. They plainly encode part of what an xG model requires, and the question we address is what proportion.

We treat this as a measurement problem rather than a feasibility question, because the practical implication depends on the magnitude. Recovery of a small fraction of a coordinate model would be of limited interest. Recovery of most of one would make xG estimable in any competition for which a commentary feed is written, a substantially wider population of competitions than those with open coordinates.

Contributions

1. We quantify how much of a coordinate xG model is recoverable from commentary text. On 8,825 shots for which both a commentary sentence and a StatsBomb coordinate xG are available, a model given only the sentence recovers 90.6% of the coordinate model's discrimination above chance (Section 5.2).
2. We calibrate that figure against the agreement observed between commercial providers, which is the only external scale available for it (Section 5.3).
3. We establish that the extraction is not the binding constraint. Models given the full sentence, including all 1–4 grams and a sentence embedding, perform worse than the eighteen extracted fields by 0.010 AUC (Section 5.4).
4. We characterise concept drift in text-derived features. The ranking is stable across a decade, replicating [9] on a different feature source, whereas the calibrated level is not. We localise the drift to two phrases, exhibit control phrases that do not move, and give a leak-free in-season correction (Section 6).
5. We release the full pipeline. Collection, parsing, estimation, evaluation and every figure reported here are reproduced by a single script [4].

2 Related Work

Coordinate-based xG. Expected goals models have been studied for roughly two decades. Recent work addresses explainability [3], player and position correction [10], and downstream validity [6]. In each case shot location is taken as given.

Data requirements. Robberechts and Davis [9] examine what data an xG model requires. They report that a basic logistic specification converges at approximately 6,000 shots, that estimation on less recent data incurs “only a negligible performance hit”, that league-specific and cross-league models “perform equally”, and that data from different providers should not be pooled. They further argue that calibration rather than ranking is the appropriate objective. Their results bear on the present work in two ways: our staleness and transfer results replicate theirs on a different feature source, and our drift result is interpretable only against the coordinate-side baseline they establish.

Language and football. Text has been used to forecast match events from event sequences [7], and to render an xG model’s coefficients as natural-language explanations of individual shots [8]. The latter proceeds in the opposite direction to the present work, mapping a fitted model onto words rather than words onto a model. The two are complementary rather than equivalent.

Prior use of commentary as input. An unpublished public repository [2] scrapes Premier League match commentary and states that shot position can be inferred from the commentary text, from which goal-conversion rates by shot category are derived. To our knowledge this is the earliest use of commentary text as a shot-quality input, and it precedes the present work. It reports no fitted model, no held-out evaluation, no comparison against a reference xG model, and no AUC, log loss or Brier score. The contribution claimed here is therefore the measurement rather than the representation.

Agreement between providers. Published comparisons of Opta, Understat, StatsBomb and Wyscout report match-level xG correlations between 0.86 and 0.96 depending on the pair [5]. We use these figures as the reference scale for our own agreement with StatsBomb in Section 5.3.

Feature sets in public models. American Soccer Analysis publish their xG specification [1], comprising distance, available goal mouth, whether the shot was headed, whether it followed a cross or through ball, and pattern of play including corners, free kicks, fast breaks and penalties, estimated by logistic regression. Setting aside the two continuous geometric terms, this is close to the field set a commentary sentence yields, which provides independent support for the choice of features in Section 4.2.

Table 1. The commentary corpus. Premier League spans 2015/16 and 2022/23–2025/26; the other five competitions cover 2022/23–2025/26.

Competition	Matches	Shots	Goals	Goal rate
Premier League	1,892	47,431	5,489	11.6%
La Liga	380	9,240	1,024	11.1%
Serie A	380	9,065	922	10.2%
Bundesliga	306	7,853	990	12.6%
Ligue 1	305	7,391	863	11.7%
Primeira Liga	306	7,000	821	11.7%
Total	**3,569**	**87,980**	**10,109**	**11.5%**

3 Data

3.1 Commentary

ESPN publish an Opta-derived English commentary feed through a public endpoint that requires no authentication. We collected 3,569 matches across six competitions (Table 1), from which 87,980 shots and 10,109 goals were parsed.

3.2 The reference model

StatsBomb release Premier League 2015/16 as open data [11], including a shot location, a freeze frame of all players in view, and their own xG estimate for each shot. Joining this release to ESPN commentary for the same fixtures is, to our knowledge, the only available means of evaluating words and coordinates on identical shots.

The join is deliberately conservative. A shot is paired only when the fixture, the team and the minute all agree, and an unmatched shot is discarded rather than imputed. This yields 8,825 shots across 373 matches. As a check on the join itself, the two sources agree on whether a shot was a goal for 99.71% of paired shots, disagreeing on 26. All evaluation in Section 5.2 takes StatsBomb's goal indicator as the label, so that their model is not penalised for annotation differences in the commentary feed.

4 Method

4.1 Outcome leakage in the commentary sentence

Commentary states the outcome of a shot before describing the chance that produced it. Every line opens with "Goal!", "Attempt missed", "Attempt saved" or "Attempt blocked", and the outcome is frequently restated in the verb, as in "is saved in the bottom left corner" or "hits the left post". That text constitutes the label. A model estimated on the unmodified sentence attains an AUC of 1.00 while learning nothing about the quality of the chance.

Table 2. Goal rate by parsed field, 37,929 Premier League shots (2022/23–2025/26). Overall rate 11.8%.

Field	% shots	Goals	Field	% shots	Goals
penalty	1.0	82.7%	from cross	18.6	10.7%
six yard box	5.0	39.6%	assisted	75.9	10.7%
after fast break	2.7	35.5%	header	18.2	10.5%
through ball	3.5	25.7%	difficult angle	2.5	8.1%
after corner	9.8	16.4%	side of box	18.5	5.6%
centre of box	36.2	14.4%	outside the box	30.6	4.1%

Reliable removal of the outcome proved difficult, and four distinct leaks survived successive attempts:

1. Stripping only the opening clause left penalties converting at 100%.
2. A blacklist of outcome words failed to remove the verbs; inspection of the fitted coefficients found `goal` weighted at +7.99.
3. Under a whitelist, retaining only spans matching a fixed set of chance-describing patterns, the phrase "from a free kick" still converted 60 of 60. Free kicks are consequently excluded from the feature set.
4. The `header` indicator was being set by the phrase "headed *pass*", mislabelling 1,636 shots and raising their apparent conversion from 10.0% to 13.8%. Fields describing the shot itself are now read only from the text preceding "assisted by".

These cases share a structure. A leak in text features is a phrasing that was not anticipated, so a fixed word list cannot bound the problem and the safeguard must be behavioural. We therefore estimate a model on the stripped text and require that it not separate goals beyond a specified threshold, which we set at AUC 0.85 against 1.00 for the unmodified text. We additionally flag any n-gram whose Wilson lower bound on conversion exceeds that of penalties, the highest-converting legitimate feature. This procedure identified leak (3), and correctly admitted a genuine high-conversion phrase, "close range following fast break" at 26 of 32, that a fixed 80% threshold had rejected.

4.2 Features

Eighteen fields are extracted by whitelist regular expression: six location zones (`six_yard`, `centre_box`, `side_box`, `outside_box`, `long_range`, `difficult_ang`), three body parts, five build-up descriptors (`from_cross`, `from_headed_pass`, `from_through`, `after_corner`, `after_break`), `assisted`, `penalty`, and the minute of the match.

Table 2 reports the conversion rate associated with each field. The observed range spans roughly a factor of twenty, from 4.1% for shots described as coming from outside the box to 82.7% for penalties.

Table 3. Words against coordinates, 8,825 shots, Premier League 2015/16.

Model	Input	AUC	Log loss	Brier
Commentary model	the commentary sentence	0.7826	0.2711	0.0764
StatsBomb	coordinates, freeze frames, 16 player positions	0.8118	0.2555	0.0715

4.3 Specification and evaluation protocol

The model is a logistic regression on standardised features. We adopt this specification rather than a more flexible learner because eighteen predominantly binary fields offer limited scope for interaction effects, a choice Section 5.4 supports empirically.

Two distinct fits are reported and should be distinguished:

- the *primary* fit, estimated on Premier League 2022/23–2024/25 (28,735 shots) and evaluated on the held-out 2025/26 season (9,194 shots);
- the *comparison* fit, estimated on Premier League 2022/23–2025/26 (37,929 shots) and evaluated on 2015/16, which is excluded from its estimation sample. This is the fit evaluated against StatsBomb.

All estimation is restricted to the Premier League, following [9] on pooling across sources. The remaining five competitions are held out entirely and used as a transfer test (Section 5.5).

5 Results

5.1 Held-out season

On the 2025/26 season, which is not used in estimation, the primary model attains an AUC of 0.7709 and a Brier score of 0.0840 against a base rate of 11.3%.

5.2 Comparison against a coordinate model

Table 3 reports the central result. Both models are evaluated on the same 8,825 shots, labelled by StatsBomb.

$$\frac{0.7826 - 0.5}{0.8118 - 0.5} = 0.906 \tag{1}$$

The commentary model therefore recovers 90.6% of the coordinate model's discrimination above chance. The two mean predictions agree to within 0.001, at 0.0972 against 0.0982, so the information the words omit does not enter as a systematic bias. Per-shot correlation is 0.735 (Spearman 0.628), with a mean absolute difference of 0.052.

Table 4. Match-level xG agreement. Provider figures from [5] (five leagues, five seasons, ≈4,290 matches); ours computed on 746 team-matches of the join.

Pair	Correlation
Opta × Understat	0.96
Opta × StatsBomb	0.92–0.93
StatsBomb × Understat	0.92–0.93
Wyscout × the others	0.86–0.88
Commentary model × StatsBomb	0.869

5.3 Interpreting the recovery figure

A recovery figure of this kind requires an external scale. The natural comparison is the agreement observed between commercial providers, all of which work from coordinates and none of which agrees exactly with the others [5]. Table 4 places the commentary model on that scale.

Mean absolute difference is 0.249 xG per team-match, with a median of 0.198 and a maximum of 1.54. Mean xG is 1.150 for the commentary model against 1.161 for StatsBomb and 1.192 observed goals.

A model estimated from English sentences therefore agrees with StatsBomb to approximately the degree that a commercial provider does. Two qualifications apply. First, the levels of aggregation are similar but not identical: the published figures are per match across five leagues and five seasons, whereas ours is per team-match on one league and one season. Second, a frequently repeated figure of approximately 1 xG mean absolute difference between providers, with a maximum of 3.88, is deliberately not used here. The attribution of 3.88 to a single club suggests a season aggregate, which would make the published figure substantially tighter per match than our 0.249 rather than looser, and the underlying source is not openly available, so the unit cannot be confirmed.

5.4 Whether extraction is the binding constraint

The residual 9.4% may arise from either of two sources: the regular expressions capture only anticipated phrasings, or the sentence does not contain the missing information. These can be distinguished by supplying models with the entire sentence and testing whether they improve on the eighteen extracted fields. Table 5 reports the comparison.

Every specification given the full text performs worse, including the sentence embedding, and appending the embedding to the extracted fields degrades those as well. The remainder of the sentence consists largely of player and team names and connective text, which the models fit as noise. The cost of reading more of the sentence is 0.010 AUC, so no more capable reader, including a large language model used as an extractor, can be expected to close the gap. The residual 9.4% corresponds to information the sentence does not encode: the exact distance within a zone, the shot angle, and the number of intervening defenders.

Table 5. Reading more of the sentence, held-out 2025/26 season.

Model	Sees	AUC
18 extracted fields, logistic *(primary)*	18 fields	0.7709
18 extracted fields, gradient-boosted trees	18 fields	0.7695
every 1–4 gram, TF-IDF	all words	0.7612
sentence embedding (MiniLM)	all words	0.7584
embedding + extracted fields	both	0.7589

Table 6. The Premier League fit applied to five further competitions without re-estimation.

Competition	Shots	Goal rate	AUC	Mean xG
Premier League *(estimation sample)*	9,194	11.3%	0.7709	0.127
La Liga	9,240	11.1%	0.7730	0.119
Ligue 1	7,391	11.7%	0.7807	0.122
Bundesliga	7,853	12.6%	0.7795	0.125
Serie A	9,065	10.2%	0.7702	0.115
Primeira Liga	7,000	11.7%	0.7871	0.122

5.5 Transfer across competitions

The claim that xG becomes estimable wherever commentary is published holds only if the model transfers. We applied the Premier League fit to five further competitions without re-estimation or tuning (Table 6).

Mean AUC across the five held-out competitions is 0.7781 against 0.7709 on the estimation league, a transfer cost of -0.0072 and hence a marginal improvement. Calibration is maintained, with mean predictions of 0.115 to 0.125 against observed rates of 10.2% to 12.6%, and the ordering across competitions is preserved. This replicates the cross-league result of [9] on a feature source they do not consider. A plausible mechanism is specific to this setting: the commentary template is constructed identically across competitions, so the feature distribution is comparatively stable across them.

6 Concept Drift in Text-Derived Features

6.1 The observed discrepancy

On the held-out season the model predicts 1.54 goals per team-match against 1.38 observed, an over-estimate of 11.9%. This decomposes as follows.

		Factor
league goal rate, 2022/23 → 2025/26	0.1199 → 0.1134	$\times\,1.057$
mean prediction vs. its own training rate	0.1199 → 0.1270	$\times\,1.059$
observed over-estimate		**1.119**

Table 7. Share of shots and conversion rate for four commentary phrases, Premier League 2015/16 against 2025/26.

	2015/16		2025/26		
Phrase	% shots	conv.	% shots	conv.	
"six yard box"	3.22	43.5%	5.96	35.8%	shifts
"following a fast break"	0.73	49.3%	3.30	28.4%	shifts
"centre of the box"	32.84	14.2%	36.44	13.2%	stable
"penalty"	0.94	83.1%	1.01	82.8%	stable

The first factor reflects a decline in the league's conversion rate. The second requires explanation. The mean prediction of a logistic regression on its own estimation sample equals the sample base rate by construction, so a mean of 0.1270 on 2025/26 implies that the parsed features of those shots are more favourable than those of the shots on which the model was estimated, while the shots themselves produced fewer goals.

6.2 Localising the shift to individual phrases

Table 7 attributes the second factor to two phrases. Over the decade separating the two seasons, "following a fast break" became approximately 4.5 times more frequent while its conversion rate fell to a little over half its former value. A change of this magnitude in the underlying football is implausible on both counts simultaneously. The more parsimonious explanation is a change in how loosely the phrase is applied by whoever writes the feed, with the consequence that a model conditioned only on the words attributes higher quality to the shot than was warranted.

The control phrases are as informative as the two that move. "Penalty" admits no interpretive latitude and is unchanged, at 0.94% of shots converting at 83.1% in 2015/16 against 1.01% at 82.8% in 2025/26. "Centre of the box" is nearly as stable. The shift is therefore not a general artefact of the parser or of corpus composition, but is confined to phrases whose application requires a judgement by the annotator.

Scored with the primary fit, the sign of the calibration error reverses across the decade, from −2.7% on 2015/16 to +11.9% on 2025/26. The estimation sample lies between the two.

6.3 Re-estimation does not correct the level

Training window	Over-estimate
all three seasons *(primary)*	+11.9%
last two seasons	+11.9%
last season only	+9.1%
recency weighted, half-life one season	+9.9%

The most favourable of these recovers 2.8 of the 11.9 percentage points at the cost of two thirds of the estimation sample. The limitation is not the age of the data. The conversion rate of the forthcoming season is not present in the current season's data under any weighting scheme, so no re-weighting of past observations can recover it.

6.4 In-season intercept correction

Once several hundred shots of a season have been played, the realised conversion rate for that season is observable, and a single additive shift on the intercept restores the level. We walk forward through 2025/26 in blocks of 500 shots, obtaining the shift by bisection on only those shots played before each block, so that no future observation enters the correction applied to it.

	mean xG	actual	over	log loss
uncorrected	0.1267	0.1132	+12.0%	0.29523
recalibrated	0.1165	0.1132	+2.9%	0.29443

The comparison covers 8,694 shots evaluated after a warm-up of 500 shots, approximately twenty matches, during which the model runs uncorrected because the realised rate is too imprecise to correct towards. An additive shift on the intercept is a monotone transformation of the predicted probabilities, so AUC is invariant by construction; the accompanying test suite asserts this. The purpose of that assertion is to prevent a subsequent adjustment to the coefficients, which would improve calibration at the expense of the induced ranking.

6.5 Relation to prior results on staleness

Robberechts and Davis [9] treat calibration as the primary objective in xG evaluation and report that staleness has a negligible effect. We reproduce their ranking result, finding that a decade-stale model costs 0.0068 AUC, while observing the predicted level move by approximately 10% over the same interval. The two findings are consistent, and taken together they separate two properties that are often reported jointly. Event-data xG is stable in precisely the respect, calibration, in which a text-derived model is not, and the instability we observe is confined to phrases requiring annotator judgement. Coordinates are measurements and do not drift; phrases are authored and do.

To our knowledge this is the first characterisation of concept drift in text-derived features for sports data. It carries a practical implication: a system conditioned on commentary phrasing requires periodic correction of the level within a season, and does not require re-estimation.

7 Limitations

- *A single reference model on a single season.* The headline comparison is against StatsBomb alone, on Premier League 2015/16. A second provider

or a second season would strengthen it materially. The availability of open releases is the binding constraint.
- *Free kicks are excluded.* The phrase “from a free kick” leaked the outcome, converting 60 of 60, and the feature was removed rather than repaired. A specification that handled free kicks correctly would likely perform better.
- *Matches without commentary are discarded without record.* A small number of fixtures return an empty feed and contribute nothing to the corpus. The rate is low but non-zero, and is not currently instrumented.
- *Latency is not measured.* We assert that commentary is published quickly enough for live estimation without evidence. A scheduled matchday observer exists, but on the day tested the cloud scheduler triggered 2 of 12 scheduled intervals and did not cover the kickoff.
- *The novelty claim rests on an incomplete search.* Our search of arXiv, OpenAlex and Crossref returned 5 of 6 recall probes, and not consistently the same five. A subsequent search of grey literature, which located [2], returned 3 of 4 probes by name, and one major discussion forum refuses automated access, which is a plausible location for unpublished work of this kind. Both searches are recorded query by query in the repository. We have not searched licensed indexes, thesis repositories, or non-English sources.
- *The provider-agreement comparison is approximate*, for the reasons given in Section 5.3.

8 Conclusion

A model estimated from commentary sentences recovers 90.6% of a coordinate expected-goals model’s discrimination above chance, and agrees with it at the team-match level to approximately the degree that one commercial provider agrees with another. The residual gap is not attributable to the extraction step, since richer readers of the same sentence perform worse; it corresponds to information the sentence does not encode. The model transfers to five further competitions without re-estimation, which we attribute to the commentary template being constructed identically across them.

The failure mode we identify lies in the calibrated level rather than the ranking. Text-derived features drift where coordinate features do not, and the drift is confined to phrases whose application requires a judgement by the annotator, while unambiguous phrases are unchanged over a decade. A single intercept, refitted within the season on shots already played, corrects the level without affecting the ranking.

For competitions with no prospect of open coordinate data, which comprise the large majority, this is the difference between an available xG estimate and none.

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Reproducibility

Every quantity reported in this paper is produced by a single script in the accompanying repository [4], which executes the pipeline in dependency order from collection through to the final tables. The repository also contains a test suite of 50 tests. These include behavioural guards against outcome leakage, a check that the walk-forward correction of Section 6 uses no observation from after the block it is applied to, an assertion that the intercept shift leaves AUC unchanged, and a dependency graph that fails if any derived artefact predates the inputs from which it was built. The literature and grey-literature searches on which the novelty claim of Section 2 rests are recorded in the repository query by query, together with the recall probes discussed in the limitations.

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